LM1 Intercorperate Investments

Investment Categories

债权及未产生大影响的股权　　　　长期股权投资

	Financial Assets	联营 Associates	合营 Joint Ventures	Business 子公司 Combination
Degree of Influence	no significant influence	significant influence	shared control	control
Percentage of Interest	< 20%	20% - 50%	varies	> 50%
Term of Investee		associate		subsidiary
Accounting Treatment	AMC FVOCI FVPL	Equity Method	Equity Method	Consolidation / Acquisition Method

只是 guideline 更看实质影响

[Percentage of ownership]

<20%	Lack of Influence
20% ~ 50%	Significant Influence
>50%	Control

direct 与 indirect 合计

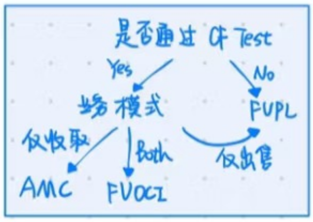
Financial Asset

❶ 分类

2原则
- Cash Flow Characteristic Test：未来CF是否基于本金及利息的合同现金流？
- Business Model Test：收取合同CF / 出售Financial Asset / Both

3分类
- AMC：通过 CF Test 且目的仅收取合同CF
- FVOCI：通过 CF Test 且目的 Both
- FVPL：其余所有（衍生品一定FVPL） 兜底.

AMC只有债券
股票一定在FVOCI or FVPL



2指定
- FVPL option：指定为FVPL以避免 accounting mismatch.（债）
- FVOCI：指定非交易性股权投资为 FVOCI（股）.

一经指定,不得改变!

❷ Measurement

只有Div Income进I/S.（R/E）.

Impairment：Expected Loss Model

	FV through P/L	FV through OCI	Amortized cost 不认unrealized G/L.
Balance sheet	• Fair value	• Fair value • Unrealized G/L ->OCI (I/S when realized) 例外：在IFRS下，投资外国债券的外汇变动计入I/S	• Amortized cost
Income statement	• Interest • Dividend • Realized or unrealized G/L 浮盈浮亏.	• Interest • Dividend • Realized G/L	• Interest • Realized G/L

平价	溢价	折价
I= coupon	I<coupon	I>coupon
A = S	A< S	A>S.

若Par Value发行
始终保持面额.

Amortized cost：
BASE法则.　　B/S中的AC
Beginning + 应收 - 实收 = Ending.
应收 = Inv×r.　实收 = Coupon

3类 Int Income
= Beginning Investment × r 都进I/S.

3 Reclassification (不重要，尽毛即可)

Financial Asset
- 股 — 不允许重分类，仍是 FVPL / 指定FVOCI
- 债 — 业务模式重大变动可重分类 ↑ 少见，不鼓励 — AMC / FVOCI / FVPL 指定FVPL不可重分类

金融负债、衍生品 — 经确认不可重分类

Reclassification Date：业务模式变更后下一个会计周期的第一天、未来适用法

Investment in Associates 联营 (associate or investee)

direct, indirect 会计，且判断时
- IFRS：current 可行权的各类 option 以股会计，记账以事实记
- GAAP：option 不算.

☆ ❶ Equity Method = one-line consolidation 单项合并

① Basic Principal：

初始：长投 at cost (FV).

Then
- investee's net income 按比例记在 investor's 长投，RE相应增长
- investee 分红理解为退款，I/S 不处理，B/S中：$\boxed{\text{Cash}+,\ 长投-}$ ∵ investee 所有者权益降低了！
- 若 investee 亏钱，长投以0为下限，其余进入 $\boxed{备查簿}$

公式：BASE 法则.

B/S：Inv in Associate = Cost + NI×% − Div%　← 追加额外溢价折旧！因为NI要调成FV

I/S：Inv Income = NI×%

② Excess Purchase Price.

PP [excess [goodwill ; FV [PP [FV appre ; BV

- goodwill → not amortized, review for impairment. 不单独列示于报表，隐含在长投中 ∴是对长投进行减值测试
- FV appre → FV−BV, Amortization of FV Appreciation. $\frac{FV-BV}{year}$
- BV → BV of net identifiable asset = Equity BV 可辨认资产与负债差额

公式：

Cost = Purchase Price = B.V + F.V appre + Goodwill

Goodwill = PP − FV%

③ Impairment of 长投

× temperary FV高时常做减值测试，一经计提减值不可返回.

IFRS：Carrying Value − Recoverable Amount (disposal value. value in use 高者).

GAAP：Carrying Value − FV

BV of Identifiable Net Asset

④ Transaction with associate.

NI调整未实现内部交易损益，实现后再加回，不重复操作

upstream 逆流：investee卖东西给investor

downstream 顺流：investor卖东西给investee

顺流、逆流一样处理，无论谁有unrealized，都减！

未出售给独立第三方/未消耗都是 unrealized!
若是损失，均认定为 realized!

$\boxed{\text{NI剔除 unrealized Profits}}$

⑤ FV option 公允价值选择权.

长投确认当天
- IFRS：部分investor机构，有权利选择
- GAAP：全部机构

Consolidation

❶ Combination

- 控股合并Aquisition：A+B=A+B → 只有控股合并 编制合并报表
  - 同控：权益结合法
  - 非同控：购买法 Acquisition Method.
- 吸收合并Merger：A+B=A
- 新设合并Combination：A+B=C

❷ Consolidation Process

① B/S合并思路：
- A公司(BV) + B公司(FV). 一定100%并
- A与A合并. L与L合并. E不合并 ⟶ A的 E + Minority Interest.
- 溢价部分单独计算 +F
  - Partial Goodwill = 收购部分Goodwill
  - Full Goodwill = Partial Goodwill ÷ 收购%
- Full⇔Partial 若不平，调 Minority Interest
  - Full Goodwill → MI = 对价 × 少%
  - Partial Goodwill → MI = FV × 少%

只有Consolidation有MI.
Equity Proportionate 无MI

② I/S合并思路.

将B公司BV调为FV.简单加总 (Inventory导致COGS调整. F/A. I/A导致 D&A调整)

Goodwill

❶ 计算：先Partial后Full

❷ Impairment.
- IFRS: BV vs. Recoverable Amount (of reporting unit include GW) 超出部分减值. 到0按%减Asset
- GAAP
  - step1: BV > FV减值.
  - step2: BV - Implied FV.

Equity Method → Consolidation.

NI不变！
- Equity ↑. ROE↓.
- Asset ↑. ROA↓.
- Sales ↑ net margin↓

Equity Method 财务指标好看

pooling vs. purchase.

ROA. ROE. Profit Margin 都更高.

Historical Book Value. A-L更小.

SPE

- IFRS：均合并 不独立 依赖母公司.
- US GAAP：
  - VIE 合并
  - not VIE 不合并 (出表).

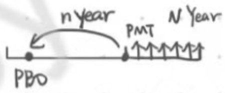
LM2. Employee Compensation

分类 DC VS. DB

看谁承担风险. DC Plan 员工承担

PBO (PUDBO) 计算

累计福利单位折现.



基于退休前工资算 PMT

❶ 2种计算方法. 退休工资 = current annual salary × $(1+g)^{\text{year to retirement}-1}$

① PMT. N. I/Y. FV=0 → PV　　$PV \div (1+r)^n = PBO$

② Annual unit credit per service year (即 养老金退休时点 PV / 总年份)

PMT (only work 1 year). N. I/Y. FV=0 → PV　　工作 x 年. PV 乘 x. 折现.

❷ ABO. 基于当前工资算 PMT.

PBO : ABO = 退休前比当前工资倍数.

B/S 核算 (Base 法则).

Funded status = Plan assets - PBO

Plan Assets
Fair value at the end of the year
= Fair value at the beginning of the year
+ Employer contributions 公司投钱
+ Actual return 投资回报
- Benefit paid to employee 公司发钱

PBO = PV(养老金)
PBO at the end of the year
= PBO at the beginning of the year
+ Current service cost +1年 + 1% = PV(Δ养老金) = unit credit 折现
+ Interest costs = PBO初 × r
+ Past service cost (plan amendments during the year) 计划修改 1% → 1.5%
+ Actuarial losses
- Actuarial gains during the year 假设改变
- Benefit paid to employee 公司实际发钱

Plan asset > PBO → Overfunded
Plan asset < PBO → Underfunded
→ 是资产时看有无超过 Ceiling.

I/S 核算. TPPC

IFRS

P/L
- Current Service Cost
- Past Service Cost
- Int Cost = PBO初 × r
- − Int Income = PA初 × r　discount rate

OCI
- Actuarial G/L 精算假设
- Actual Return − Int Income

} Remeasurement

US GAAP

P/L
- Current Service Cost
- Int Cost = PBO初 × r
- − Expected Return = Asset × E(R).
- Amortization of { Past Service Cost / Actuarial G/L }　根据剩余工作年限摊销.

OCI　Unamortized { Past Service Cost / Actuarial G/L }

假设条件变化. r / g / E(r) 方向判断即可.

❶ r↑ { B/S: PBO↓. Funded Status↑ ; I/S: C.S.C↓. Int Cost = PBO初 × r ↓. (↓大 ↑小) }

❷ g↓ → 退休前工资↓ { B/S: PBO↓ F.S↑ ; I/S: C.S.C↓. Int Cost↓. }

❸ E(r)↑ US GAAP { I/S: Pension Expense↓ ; B/S: PBO. F.S unchanged. }

Assumption	Impact on Balance Sheet	Impact on Periodic Cost in Income Statements	Impact on Total Periodic Pension Cost	Conservative or aggressive assumption
Higher Discount Rate	Lower obligation	Periodic pension costs will typically be lower, because of lower opening obligation and lower service costs	Lower TPPC, because of higher funded status	Aggressive
Lower Rate of Compensation Growth	Lower obligation	Lower periodic pension expense, because of lower service costs	Lower TPPC, because of higher funded status	Aggressive
Higher Expected Return on Plan Assets	No effect, because fair value of plan assets is used on balance sheet	Not applicable for IFRS; Lower periodic pension expense under US GAAP	No effect, because no effects on funded status	Aggressive under US GAAP

Interest expenses = PBO × r
- Int ↓, if the employee is young
- Int ↑, if the plan is mature　PBO下降.

Total Periodic Pension Costs

TPPC：I/S Cost + OCI Cost，便于比较不同准则，IFRS. US GAAP TPPC 相等

TPPC = PPC in I/S + PPC in OCI = ΔF.S − Contribution

期初 FS + Contribution − TPPC = 期末FS

会计调整

❶ I/S Pension Cost 重分类

将 Pension Expense 部分移到 non-operating

US GAAP PPC I/S 调整：

1. + PC.养老金费用
2. − CSC (operating expense)
3. − Interest Expense (同其余 Interest Expense 合并，EBIT后)
4. + Actual Return，无视E(R)，移到 non-operating (other income)

调整

原本

I/S
- + C.S.C → SG&A 仅有CSC留在 operating
- + Int Cost → Int Expense.
- − E(R) → Int Income
- Amor of P.S.C, Amor of A.G/L } non-operating G/L

❷ CF 调整. 100% CFO 不合理.

CFF 调动相应反向变动 CFO：

缺口：Contribution < TPPC. 理解为欠钱，借钱. → $\boxed{\text{CFF + / CFO −}}$

打款不足

➢ **Under-contribution (contribution < TPPC) : "Borrowing"**

✓ CFF adjustment = CFF + (TPPC – Contribution) × (1 – tax rate) !

✓ CFO adjustment = CFO – (TPPC – Contribution) × (1 – tax rate)

➢ **Over-contribution (contribution > TPPC) : "Repayment"**

✓ CFF adjustment = CFF – (Contribution – TPPC) × (1 – tax rate)

✓ CFO adjustment = CFO + (Contribution – TPPC) × (1 – tax rate)

other post-employment benefits 处理类似于DB. 更复杂的假设. 假设同 OPB的披露.

Ultimate healthcare trend rate: 长远的小g

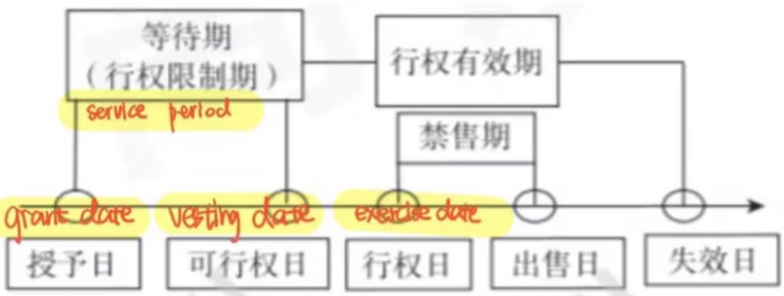
股份支付

权结 equity-settled ： stock / stock option

option变动 → cost变动 → NI变动

cost在 vesting period 摊销.

现结 cash-settled ： stock appreciation rights / phantom shares 模拟股票.



LM3. 跨国.

Transaction 外币交易会计处理

Type of Exposure

外币交易的exposure: 时间差导致. 期间汇率波动
- import purchase: 希望本币升值. 外币贬值. Liability
- export sale: 希望外币升值. 本币贬值. Asset.

- 未跨年: 应收, RE → Cash, RE. 计Realized G/L
- 跨年: Unrealized G/L 先计在第一年报表. Settle时与年末比, 计Realized G/L.

Translation 外币报表折算

Functional Currency: 关注子公司的Functional Currency. 销售. 支出、融资等作为判断依据.

① 判断方法.

	Current Rate Method	Temporal Method	CR + T Method
Parent	Presentation Currency	Presentation Currency	Presentation Currency
Control ↓	↑ Translate		↑
	Functional Currency	Functional Currency	Functional Currency
		↑ Remeasure	↑
Subsidiary	Local Currency	Local Currency	Local Currency

② 折算

- integrated subsidiary FC=RC Temporal
- independent subsidiary FC=LC Current R

Current Rate Method

When the current rate method is used, the cumulative translation adjustment needed to keep the translated balance sheet in balance is reported as a separate component of stockholders' equity

高顿教育 GOLDEN EDUCATION

①

Sales Revenue	Average Rate
(COGS)	Average Rate
(SG&A)	Average Rate
Operating Income	
Other Income	Average Rate
EBIT	
(Interest)	Average Rate
(Tax)	Average Rate
Net Income	A

② NI – Dividend = ΔRE

A (NI_{LC} × Average Rate) − Div_{LC} × Historical Rate I = B

③

Asset 1	Current Rate	Liability 1	Current Rate
Asset 2	Current Rate	Liability 2	Current Rate
		Total Liabilities	
		Capital	Historical Rate II
		Additional Paid-In	Historical Rate III
		Retained Earnings	$RE_{beg,PC}$ + B
		Cumulative Translation Adjustment	C
Total Assets		Total Equities	

CTA to balance

先I/S, 后B/S
① I/S 折算: Average Rate → NI
② B/S 折算: Asset: current rate
Liability: current rate.
= Equity { Capital: historical; R/E; OCI
③ R/E末 = R/E初 + NI − Div
④ E − Capital − R/E = OCI

Temporal Method

先B/S, 后I/S.

inventory & COGS 可能用 weighted average exchange rate, 会与之相关

①

M 确定	Cash	Current Rate	M: Accounts Payable	Current Rate
	Accounts Receivable	Current Rate	Long-Term Debt	Current Rate
Non M 不确定	Inventory	Historical Rate I	Total Liabilities	Sum (↑)
	Trading Securities	Current Rate	Capital	Historical Rate IV
	PPE	Historical Rate II	Additional Paid-In	Historical Rate V
	Intangible Assets	Historical Rate III	Retained Earnings ②	A
	Total Assets	Sum (↑)	Total Equities	Asset − Liability

R/E = E − Capital → to balance

$RE_{end} − RE_{beg}$ = ΔRE

A − $RE_{beg,PC}$ = B

⑤ RE末 = RE初 + NI − Div

ΔRE + Dividend = NI

B + Div_{LC} × Historical Rate VI = C

* Assume $DC_{presentation\ currency}/FC_{local\ currency}$

③

Sales Revenue	Average Rate
(COGS)	Historical Rate I'
(SG&A)	Average Rate
(Depreciation)	Historical Rate II
(Amortization)	Historical Rate III
Operating Income	
Other Income	Average Rate
EBIT	
(Interest)	Average Rate
(Tax)	Average Rate
⑥ 为NI之差 Remeasurement Gains and Losses	D
④ Net Income	C

to balance 5

两种方法对比

Current Rate Method	Temporal Method
All assets and liabilities are translated by current rate	"Monetary assets/liabilities" and "non-monetary assets/liabilities measured on the balance sheet date at current value" are translated by current rate
All revenues and expenses are translated by average rate	Expenses related with non-monetary items are translated by historical rate: only COGS、Dep、Amo
FIRST income statement **THEN** balance sheet	**FIRST** balance sheet **THEN** income statement
$CTA_0 + \Delta CTA = CTA_1$ "**Cumulative Translation Adjustment**" in the shareholders' equity 存量	"**Remeasurement gains and losses**" in the income statement 增量

③ 判断 T. G/L

Temporal: Exposure = M.A − M.L. ∵ Non-M historical rate 不受影响.

Current: Exposure = T.A − T.L = Equity.

>0: Asset ↑G ↓L
<0: Liability ↓G ↑L

④ T vs C 对指标的影响

① 先判断 3个 rate 大小关系

② Ratio 分子、分母用 Rate 表示.

外币贬值

LC depreciation (H>A>C)	Temporal	Current rate
Current ratio	Higher	Lower
Quick ratio CA-Inventory / C.L	Same	Same
A/R turnover Sales / A/R	Same	Same
Inventory turnover (LIFO FIFO uncertain)	Uncertain	Uncertain
Fixed asset turnover Sales / FA	Lower	Higher
Gross profit margin Sales-COGS / Sales	Lower	Higher
Net profit margin, ROE, ROA (Translation gain/loss uncertain)	Uncertain	Uncertain
Interest coverage EBIT / Int exp	Lower	Higher
LTD-to-total capital LT Debt / D+E	Lower (equity used mixed rate)	Higher

⑤ 披露
- 外汇影响NI时，translation、transaction 可分开、可合并披露
- 产生 ΔCTA 时.

⑥ 子公司恶性通胀.

标准: $(1+g)^3 - 1 > 100\% \rightarrow g > 26\%$

IFRS: ① restate for inflation 重述报表，使购买力保持一致. (Non-M. Equity. I/S items)

B/S: Monetary items 金额固定，不重述
Restate: $\times (P_1 / P_0)$. 最后倒算 R/E

I/S: Net G/L in purchasing power to balance，本质是 M.A 与 M.L 的角逐.
$\times (P_1 / P_{average})$

② × Current Exchange Rate

GAAP: ① 用 Temporal Method (一定不重述报表!)

抗通胀: Asset 存海外资产及 Non-M Asset.

LM4. Financial Institutions

Introduction

① Types

② Features { Systemic Importance. / Heavily Regulated. / Assets and Liabilities

Banks

☆ CAMELS Approach

① Capital Adequacy = $\frac{\text{Capital}}{\text{Risk Weighted Assets}}$

Total Capital	Tier 1 Capital	Common Equity Tier 1 Capital	• common stock • issuance surplus related to common stock • retained earnings • accumulated other comprehensive income • certain adjustments including the deduction of intangible assets and deferred tax assets
		Other Tier 1 Capital	other types of instruments issued by the bank that meet certain criteria
	Tier 2 Capital		instruments that are subordinate to depositors and to general creditors of the bank, have an original minimum maturity of five years, and meet certain other requirements (用途无明确限制)

整体 ≥ 8%
一级 ≥ 6%
核心一级 ≥ 4.5%

Basel Accord Ⅲ 3大亮点 {
Minimum Capital Requirement.
Minimum Liquidity: LCR持3月≥100%
Stable Funding: NSFR持一年≥100%

Liquidity Coverage Ratio $= \frac{\text{highly liquid assets}}{\text{expected CF}}$

Net Stable Funding Ratio $= \frac{\text{available stable funding}}{\text{required stable funding}}$

② Asset Quality: 定性看 creditworthiness; 定量看 expected loan loss.

③ Management Capabilities

④ Earnings | ① 银行3大收入 { 净利息差，越稳定越好 / 服务 earning / trading } 相对 stable　ROE { unbiased / sustainable / sufficient.
| ② Fair Value Hierarchy

⑤ Liquidity Position: 定量 LCR≥1, NSFR≥1; 定性: Concentration, Mismatch

⑥ Sensitivity to Market Risk: Interest Rate Risk { hypothetical: $(A-L)\times r$ = profit / reality: mismatch → 利差↑

Insurance

① P&C 财险: 更保守，投资的 liquidity 更好
营业模式: 直销 — 公司人卖
代销 — 外部人卖

5 Ratios →

- ✓ **Loss and loss adjustment expense ratio** = (Loss expense + Loss adjustment expense)/Net premiums earned.
- ✓ **Underwriting expense ratio** = Underwriting expense/Net premiums written.
- ✓ **Combined ratio** = Loss and loss adjustment expense ratio + Underwriting expense ratio. <100%
- ✓ Dividends to policyholders (shareholders) ratio = Dividends to policyholders (shareholders)/Net premiums earned.
- ✓ Combined ratio after dividends = Combined ratio + Dividends to policyholders (shareholders) ratio.

② L&I 寿险: 可预测，相对更激进. Liquidity 要求低.
营业模式: 直销 — 公司
代销 — 个人

	P&C Insurers	L&H Insurers
Contract Duration	Short term	Long term
Variability of Claims	Claims are more variable	Claims are more predictable
Business Profile	Protect against loss or damage to property	Provide benefit upon death and a savings vehicle
Investment Returns	Conservative	Seek higher returns
Liquidity	Requires a high degree of liquidity	Liquidity was less important to life insurers
Capitalization	No global standard exists	Lower capital requirements

LM1 Financial Statement Modeling (Sales Based)

高度依赖 Revenue 的预测.

I/S 预测

方法 Bottom up / Top down / Hybrid

预测

- Revenue
  - Top down: GDP. market. market share 3个 growth 叠加.
  - Bottom up:
    - Time series: $Revenue_t = \alpha + \beta \cdot Revenue_{t-1} + \varepsilon$.
    - Return on capital 更多是金融业
    - Capacity-based measure 更多是零售行业. 单店模式
- − COGS — COGS = (1 − Gross Margin) × (Future Revenue) 由 gross margin 倒推.
  - 更精确: 考虑到规模效应调高 Gross Margin; 可分 business 加总.
- Gross Profit
- − SG&A
  - Fixed Component: 保持不变 or 根据预算
  - Variable Component: 与 COGS 同比
- Operating Income (EBIT)
- − Interest Expense — Net Debt × % or Interest Expense − Interest Income 是净值!
- EBT
- − Tax Expense — EBT × Effective Tax Rate %
- Net Income

B/S 预测.

1. WC Items — Inventory Turnover $= \frac{COGS}{(Inv_{初} + Inv_{末})/2}$ 等各种 turnover 来决定
2. LT Items — 投融资基于 budget.
3. Equity
   - Capital: 增发 / 回购影响.
   - R/E: BASE 法则
   - OCI

Return.

$$ROIC = \frac{EBIT(1-t)}{Invested\ Capital} = \frac{NOPAT}{Invested\ Capital\ (D+E)}$$

税率不同时.

$$ROCE = \frac{EBIT}{Capital\ Employed}$$

Porter's Five Forces

Inflation & Deflation

price elasticity of product. 越低越不受影响.

LT Forecasting

- Normalized Earnings.
- 经济周期.
- 预测期与投资期. 先精确, 再永续. TV
  - Historical Multiple Based Approach
  - DCF